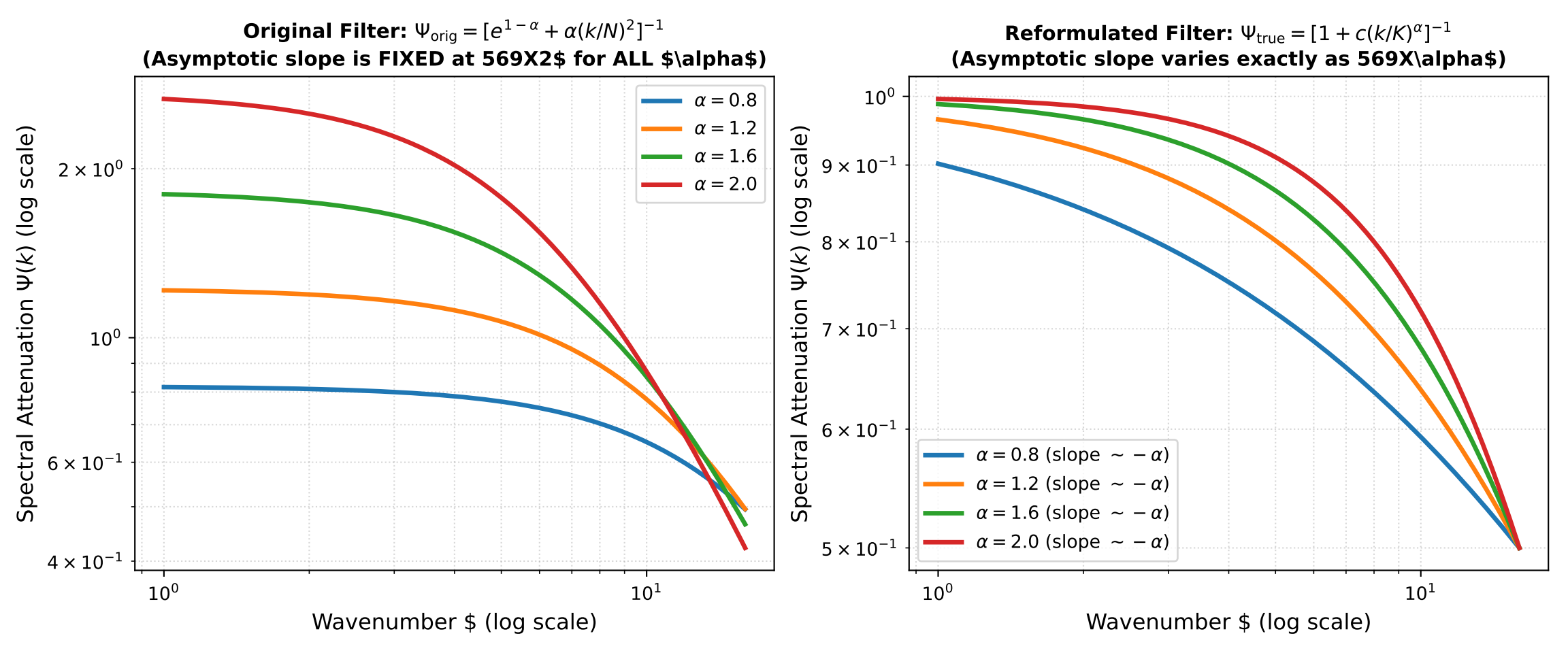
Original Filter: $\Psi_{orig} = [e^{1-\alpha} + \alpha(k/N)^2]^{-1}$
(Asymptotic slope is FIXED at 569X2$ for ALL α)
Spectral Attenuation $\Psi(k)$ (log scale)
Wavenumber $ (log scale)
$\alpha = 0.8$
$\alpha = 1.2$
$\alpha = 1.6$
$\alpha = 2.0$
2×10^0
10^0
6×10^{-1}
4×10^{-1}
10^0
10^1
Reformulated Filter: $\Psi_{true} = [1 + c(k/K)^{\alpha}]^{-1}$
(Asymptotic slope varies exactly as 569X\alpha$)
Spectral Attenuation $\Psi(k)$ (log scale)
Wavenumber $ (log scale)
$\alpha = 0.8$ (slope $\sim -\alpha$)
$\alpha = 1.2$ (slope $\sim -\alpha$)
$\alpha = 1.6$ (slope $\sim -\alpha$)
$\alpha = 2.0$ (slope $\sim -\alpha$)
10^0
9×10^{-1}
8×10^{-1}
7×10^{-1}
6×10^{-1}
5×10^{-1}
10^0
10^1